Shahjalal University of Science and Technology

Department of Electrical and Electronic Engineering

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Experiment 02

Familiarization with Basic Laboratory Instruments: Power Supply, Trainer Board, Multimeters, Oscilloscopes, and Function Generator

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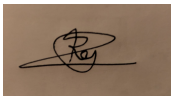
Declaration

I hereby declare that the work presented in this laboratory report entitled **"Familiarization with Basic Laboratory Instruments: Power Supply, Trainer Board, Multimeters, Oscilloscopes, and Function Generator"** is an authentic record of my own work carried out for the course **Basic Electrical and Electronic Circuit (EEE 0713 1206B)**.

This report has been prepared by me independently and has not been submitted elsewhere for the award of any degree, diploma, or credit.

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Date: August 25, 2026



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Objective

- To gain practical knowledge of basic laboratory equipment used across electrical and electronic experimentation.
- To understand operational principles and hands-on usage of regulated DC power supplies and trainer boards.
- To examine needle-deflection mechanisms and scale readings of traditional analog meters.
- To compare analog VOMs with digital multimeters for voltage, current, and resistance testing.
- To acquire operational skills in displaying, measuring, and analyzing periodic signals on analog and digital storage oscilloscopes.
- To learn signal generation procedures using function generators across different wave types.

Introduction

Laboratory testing relies heavily on the correct and cautious operation of standard instruments: power supplies to energize target circuits, breadboards or trainer panels for rapid assembly, multimeters for checking branch parameters, and oscilloscopes paired with function generators for wave observation. Mishandling these units—such as placing an ammeter across active power nodes or grounding an oscilloscope probe to a live potential—can cause hardware damage or component burnout. This experiment introduces key front-panel controls, safety steps, and basic signal evaluation procedures.

Theory

Digital/Analog Trainer Board

A trainer board combines a central breadboard workspace with integrated modules: fixed and variable DC rails, switches for logic inputs, LED indicators for output monitoring, and clock pulse generators.

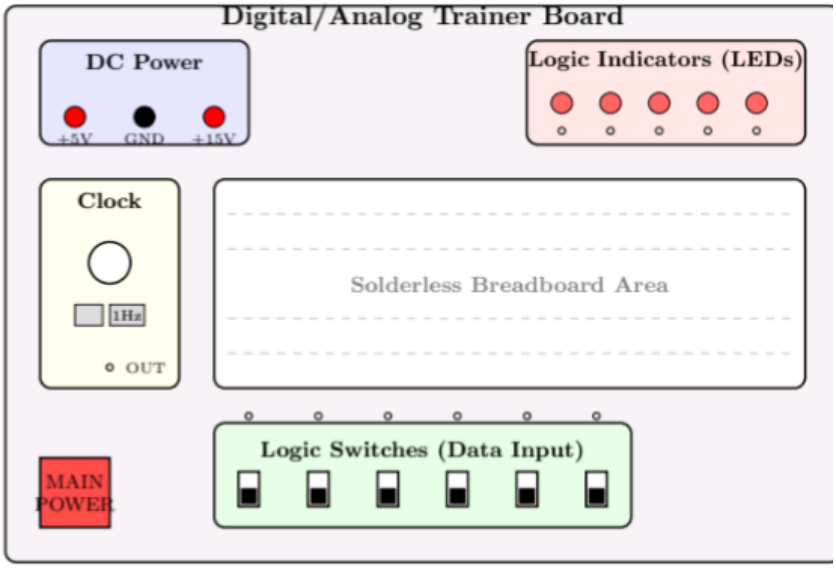


Figure 1: Standard Laboratory Trainer Board

DC Power Supply

A benchtop DC supply transforms AC mains power into a steady, adjustable DC output. In Constant Voltage (CV) mode, the equipment keeps a fixed target voltage while output current adjusts according to load resistance. In Constant Current (CC) mode, output current is clamped to a set limit.

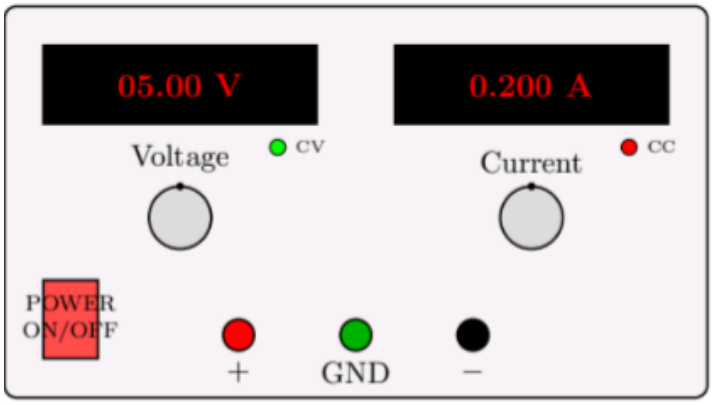


Figure 2: Regulated DC Power Supply

Function Generator

A function generator supplies standard periodic signals—sine, square, triangle, and ramp waves—with controllable frequency, amplitude, and DC offset. Benchtop generators are calibrated for a 50 Ω load.



Figure 3: Laboratory Function Generator

Analog Meters

Most analog meters rely on Permanent Magnet Moving Coil (PMMC) movements, where current passing through a coil generates torque to deflect an indicator needle.

A voltmeter pairs the coil with a high series resistor, resulting in high internal resistance so it draws negligible current when connected in parallel across a component.

An ammeter adds a low-resistance shunt in parallel with the coil, yielding minimal internal resistance; it must be wired in series with the load.

A galvanometer is a high-sensitivity center-zero meter used to detect tiny current flows and directional changes, often for bridge balancing.



Figure 4: Analogue Voltmeter



Figure 5: Analogue Ammeter

An ohmmeter utilizes an internal source to push current through an unpowered resistor and must be zero-adjusted prior to taking measurements.

An analogue frequency meter directly measures and displays the frequency of periodic AC electrical signals.



Figure 6: Analogue Galvanometer



Figure 7: Analogue Ohmmeter

Multimeters

An analog multimeter (VOM) combines voltage, current, and resistance measurement functions into a single PMMC assembly using a rotary switch. Range selection must begin at the highest setting and work downward while observing correct DC polarity.

A Digital Multimeter (DMM) replaces analog coils with solid-state analog-to-digital converters, eliminating parallax error while providing high input impedance (10 MΩ+), auto-ranging, and diode



Figure 8: Analogue Frequency Meter

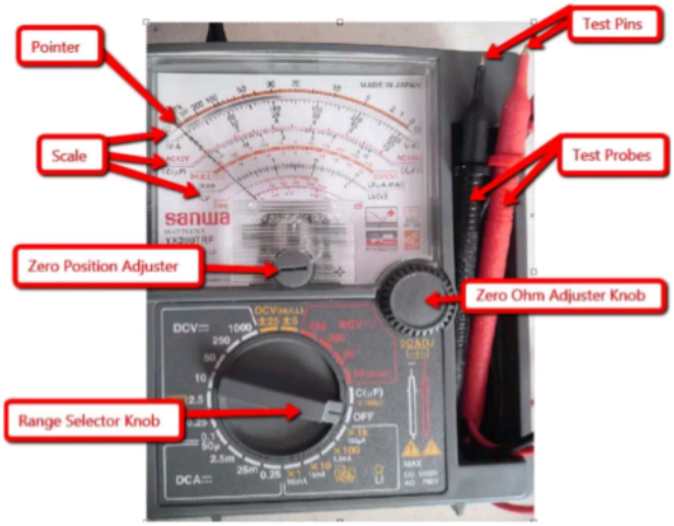


Figure 9: Analogue Multimeter (VOM)

test features.

Oscilloscopes

An oscilloscope plots voltage on the vertical (Y) axis against time on the horizontal (X) axis. Analog oscilloscopes sweep an electron beam across a phosphor CRT screen in real time.

Digital Storage Oscilloscopes (DSOs) digitize the input signal through an ADC and save it in memory, enabling single-shot capture, automated signal measurements (V_{pp}, period, frequency), and math options.

Apparatus and Components

Circuit Diagram



Figure 10: Digital Multimeter (DMM)

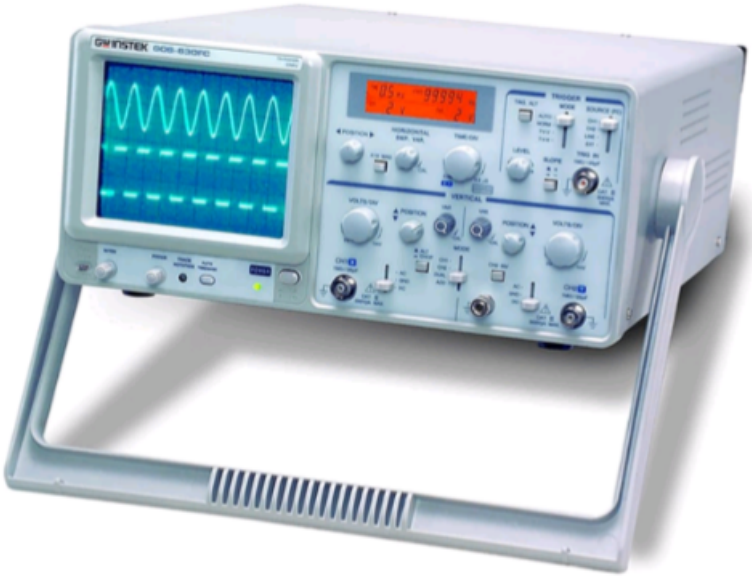


Figure 11: Analogue Cathode Ray Oscilloscope

Experimental Procedure

1. **Power Supply Setup:** Turn on the DC power supply before attaching external leads, adjusting voltage knobs to set the required level. Establish a safety current limit by temporarily shorting output terminals and turning the current dial to the target value before removing the short connection.

2. **DMM Resistance Measurement:** Switch the DMM to resistance mode and measure the 1.0 kΩ test load, cross-checking against color band calculations.

3. **Function Generator and Oscilloscope Setup:** Construct the circuit shown in Figure 13. Set the function generator to produce a 1 kHz sine wave at a designated amplitude prior to turning on output. Attach the scope probe across the resistor and adjust Volts/Div, Time/Div, and Trigger Level controls until a clear waveform appears.



Figure 12: Digital Storage Oscilloscope (DSO)

Table 1: List of Equipment Used

Item Description	Specification / Rating	Quantity
Digital/Analog Trainer Board	Built-in 5 V supply, logic switches, LEDs	1
DC Power Supply	0 V–30 V, dual output	1
Function Generator	Sine/Square/Triangular, 0 Hz–1 MHz	1
Analog Voltmeter	Benchtop voltage meter	1
Analog Ammeter	Benchtop current meter	1
Analog Galvanometer	Center-zero high sensitivity	1
Analog Ohmmeter	Benchtop resistance meter	1
Analog Frequency Meter	Panel frequency meter	1
Analog Multimeter (VOM)	Rotary range selector	1
Digital Multimeter (DMM)	Auto-ranging	1
Analog Oscilloscope	Dual channel CRT	1
Digital Storage Oscilloscope (DSO)	Dual channel LCD	1
Fixed Resistor (Test Load)	1.0 kΩ, 0.25 W	1
Solderless Breadboard	Standard Tie-Point	1
Connecting Wires & BNC Probes	Jumper wires, oscilloscope probes	As required

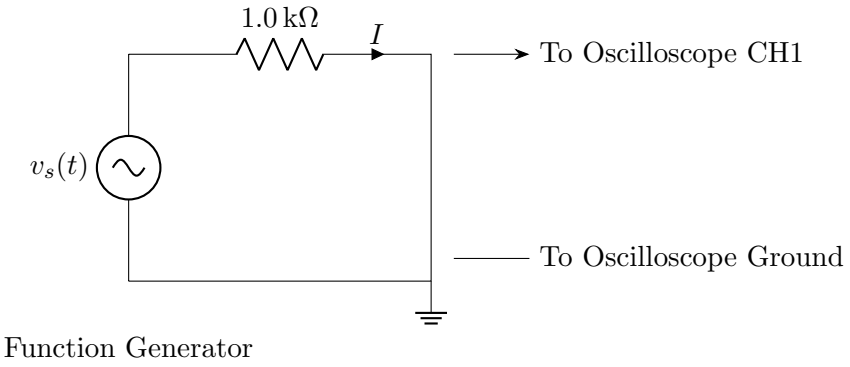


Figure 13: Basic instrument test setup: a function generator drives a resistive load while an oscilloscope probe monitors the voltage across it.

4. **Probe Compensation:** Connect the oscilloscope probe to the calibration output pin and adjust the probe's trimmer capacitor until square wave edges show square corners without overshoot.

5. **Waveform Measurement:** Calculate peak-to-peak voltage V_{pp} and period T from the display

screen, calculating frequency $f = 1/T$ to compare against generator settings.

Observations

This session served as an instrument orientation exercise, so formal signed data sheets were not compiled. Practical outcomes from testing procedures in Section 5 are summarized below:

- Constant Voltage and Constant Current operations were verified on the DC supply: the CV indicator remained active under nominal loads, switching to CC mode when load current hit the current ceiling.
- Resistance values measured on the DMM for the 1.0 kΩ load agreed with color code calculations within component tolerance margins.
- The oscilloscope showed a stable waveform once trigger settings were dialed in, confirming correct probe ground placement.
- Probe calibration produced a flat-topped square wave after trimming, ensuring accurate high-frequency signal reproduction.
- Calculated signal frequency derived from measured period T matched the generator's front-panel frequency readout.

Discussion

Because this session prioritized instrument handling over numerical calculations, error analysis was not conducted. Key practical insights include: (i) ammeters feature near-zero internal resistance and must only be wired in series to prevent short-circuiting power sources; (ii) oscilloscope ground leads connect to earth ground and must never touch active circuit nodes; (iii) function generator output readings depend on load matching, where High-Z termination doubles expected output voltages unless generator profile settings are updated; and (iv) checking probe compensation prior to taking high-frequency signal measurements prevents waveform distortion errors.

Conclusion

This lab session provided practical experience operating essential testing instruments: DC power supplies, trainer boards, digital multimeters, oscilloscopes, and function generators. Voltage and current limiting functions, series and parallel meter connections, and oscilloscope triggering procedures were systematically tested. These practical competencies are vital for subsequent laboratory work in circuit analysis.

References

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